

Lecture 9: From C to the ISA

CMPE I20: Computer Organization & Architecture

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Logistics

- HW1 is due tonight (!) at 11:59pm PT
- HW1 Quiz next Tuesday [in class](#)
 - 15min

What does a C program compile to?

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#include <stdio.h>

int main() {
    printf("Hello world!");
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- Assembly!

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.LC0:
    .string "Hello world!"
"main":
    push    rbp
    mov     rbp, rsp
    mov     edi, OFFSET
FLAT:.LC0
    mov     eax, 0
    call    "printf"
    mov     eax, 0
    pop     rbp
    ret
```

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ARM

```
main:  
    stp     x29, x30, [sp, #-16]!  
    mov     x29, sp  
    adrp    x0, .L.str  
    add     x0, x0, :lo12:.L.str  
    bl      printf  
    mov     w0, wzr  
    ldp     x29, x30, [sp], #16  
    ret  
  
.L.str:  
    .asciz  "Hello world!"
```

RISC-V

```
.LC0:  
    .string "Hello world!"  
main:  
    addi    sp, sp, -16  
    sd      ra, 8(sp)  
    sd      s0, 0(sp)  
    addi    s0, sp, 16  
    lla     a0, .LC0  
    call    printf@plt  
    li      a5, 0  
    mv      a0, a5  
    ld      ra, 8(sp)  
    ld      s0, 0(sp)  
    addi    sp, sp, 16  
    jr      ra
```

Where does assembly come from?

- The assembly language is defined by an [instruction set architecture \(ISA\)](#)

Where does assembly come from?

- The assembly language is defined by an **instruction set architecture (ISA)**
- Recall from Lecture 1:

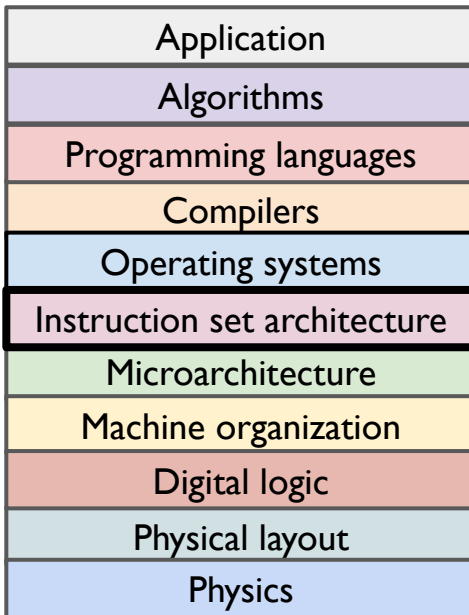
Computer architecture = **Instruction set architecture** + **Machine organization**

What is an Instruction Set Architecture (ISA)?

- An ISA describes an abstraction of a computational machine
 - Describes the “the agreed-upon interface between all the software that runs on the machine and the hardware that executes it.”

What is an Instruction Set Architecture (ISA)?

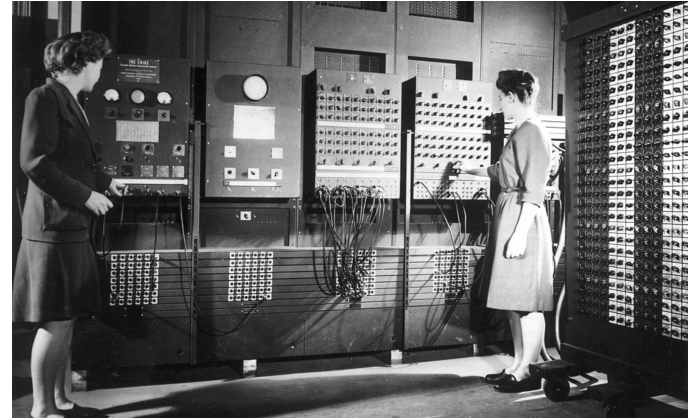
- An ISA describes an **abstraction** of a **computational machine**
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Why have an **Instruction Set Architecture (ISA)**?

In the beginning...

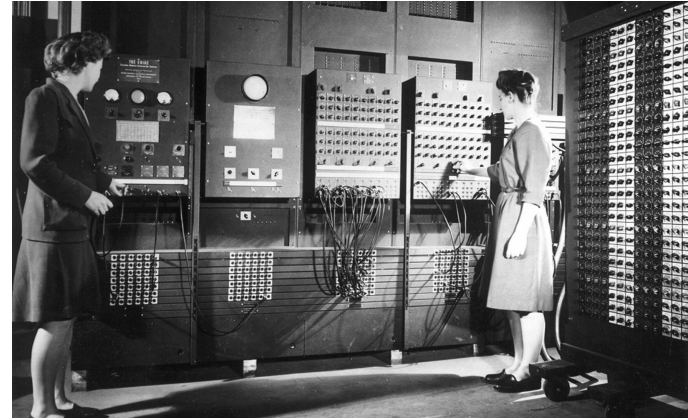
- The first computers used **physical configuration** as their “program”
- Technicians would **physically reconfigure** the machine.
 - Slow!



Why have an **Instruction Set Architecture (ISA)**?

In the beginning...

- The first computers used **physical configuration** as their “program”
- Technicians would **physically reconfigure** the machine.
 - Slow!
- Didn't let us **use the computer as a tool** for programming



Why have an **Instruction Set Architecture (ISA)**?

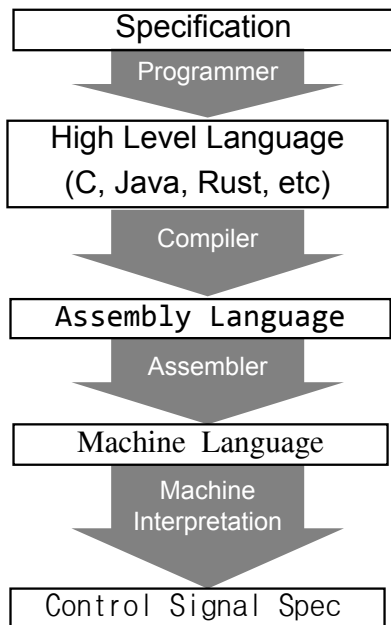
- To use the computer as a tool for programming!

Why have an **Instruction Set Architecture (ISA)**?

- To use the computer as a tool for programming!
- Idea: The **stored-program computer** where the computer comprises
 - A **central processing unit (CPU)** that
 - Executes instructions
 - Holds a small number of values in a “register file”
 - A **memory** that holds data and instructions

Why have an **Instruction Set Architecture (ISA)**?

- Computers **do not** speak English or any programming language (C, Java, etc.)



“Swap two array elements.”

```
int temp = array[index];  
array[index] = array[index + 1];  
array[index + 1] = temp;
```

```
lw  $15, 0($2)  
lw  $16, 4($2)  
sw  $16, 0($2)  
sw  $15, 4($2)
```

```
10001100011000100000000000000000  
1000110011110010000000000000100  
10101100111100100000000000000000  
1010110001100010000000000000100
```

```
ALUOP[0:3] <= InstReg[9:11] &  
MASK
```

The Instruction Set Architecture (ISA)

- The ISA is the **interface** between software and hardware

The Instruction Set Architecture (ISA)

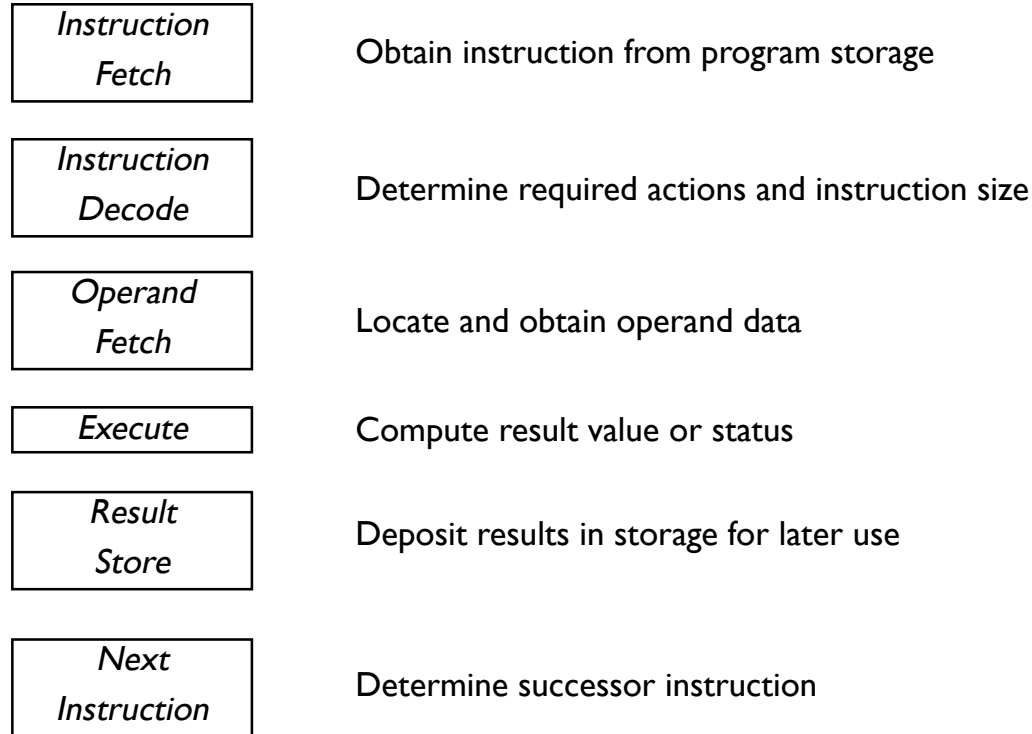
- The ISA is the **interface** between software and hardware
- The part of the architecture that is **visible** to the **programmer**
 - available **instructions** (“opcodes”)
 - number and types of **registers**
 - instruction **formats**
 - **storage** access, **addressing** modes
 - exceptional conditions

How does a computer execute instructions?

- Determined by the machine organization

How does a computer **execute instructions**?

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Preview of machine organization concepts

- **Cycle**: the smallest unit of time in a processor

Preview of machine organization concepts

- **Cycle**: the smallest unit of time in a processor
 - Processors have a **clock**
 - Cycle = a "tick"
 - Cycle time a.k.a **clock speed** = how long a "tick" takes

How to determine the clock speed of my processor?

- Find out your chip's clock rate (a.k.a frequency)
- My Macbook Air has an M2 chip with a max clock rate of 3.49 GHz
 - 1 GHz means 1 cycle per nanosecond

How to determine the clock speed of my processor?

- The M5 chip has a max clock rate of 4.6 GHz. What is its cycle time?

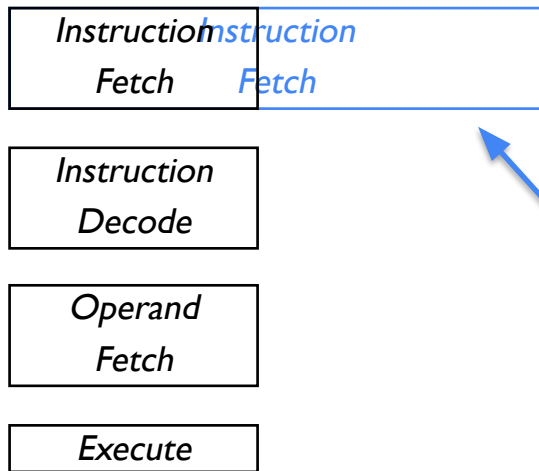
- A. 4.6 ns
- B. 0.46 ns
- C. 217 ns
- D. 0.217 ns
- E. None of the above

Given a cycle...

- Notice: Clock speed != performance !!!
- More important: How many instructions can execute per cycle?
 - Also known as the instruction rate
 - Depends on the architecture!

Preview: architecture design decisions

- Maximize **parallelism**



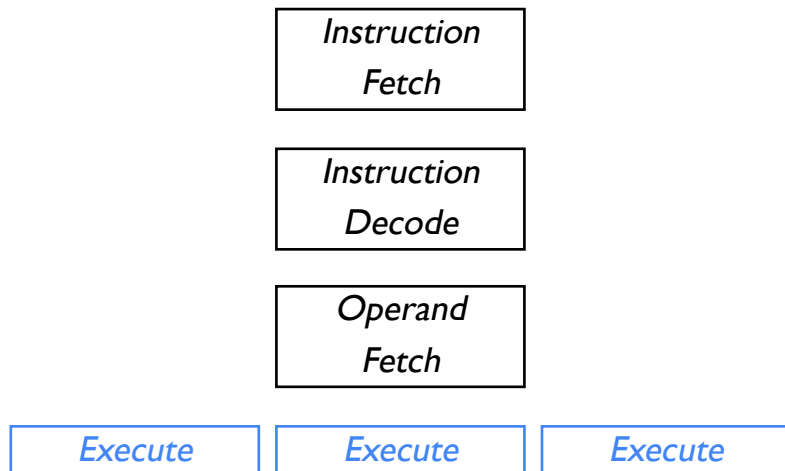
Real-world example

ARM's Thumb instruction set is (mostly)
16-bit instructions on a 32-bit machine

ISA design makes fetch “freely parallel”

Preview: architecture design decisions

- A **superscalar** processor



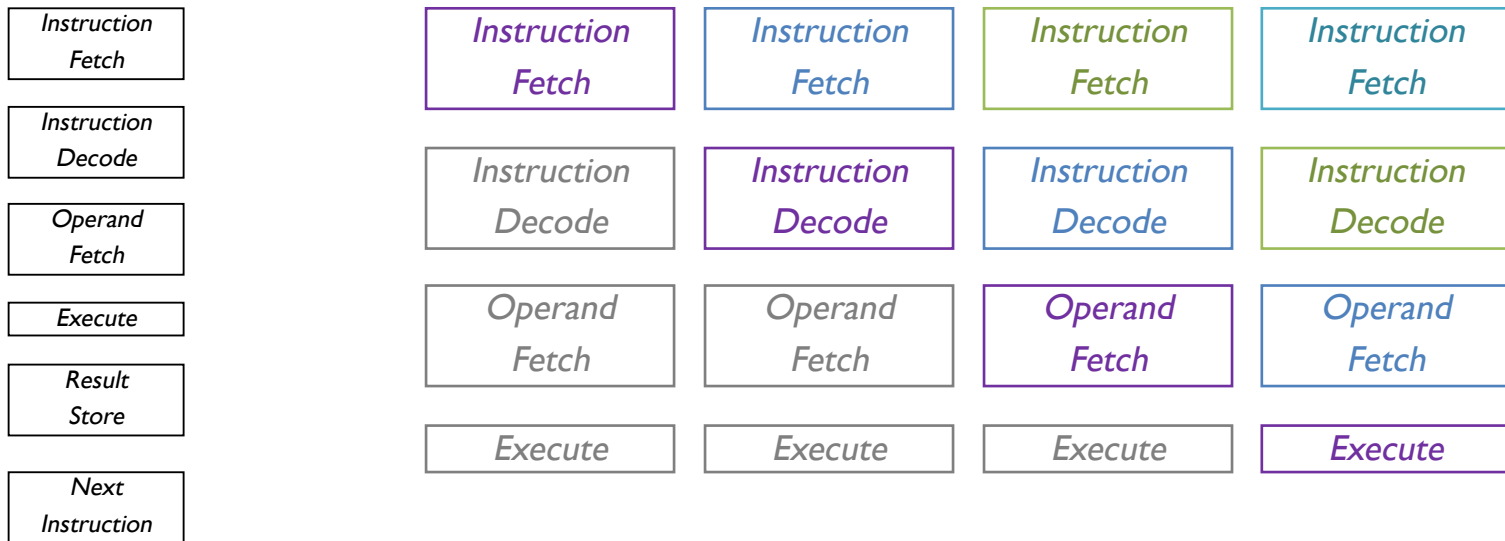
Duplication is easy but **expensive...**

How to do parallelism well?

- Learn later in class

Preview: architecture design decisions

- **Pipelining:** overlap parts of a large task to **increase throughput** w/o decreasing latency
 - **Key insight:** The less work you do in one step, the faster each step can finish



How to design an ISA?

- Goal: Want an ISA that can be designed and executed efficiently in hardware

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- Goal: Want an ISA that can be designed and executed efficiently in hardware
- By efficiency, we usually focus on
 - Latency
 - Throughput
 - Resource consumption
 - Power consumption

There are always tradeoffs!

How to design an ISA?

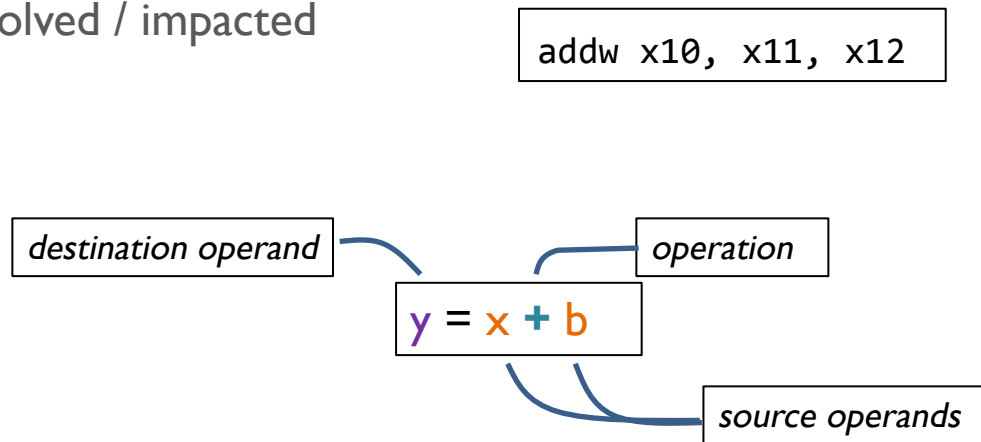
We need

- **operations**: what an instruction does
- **operands**: specify what data is involved / impacted
- **instruction format**

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How to design an ISA?

- operations

- how many?
- which ones?

- operands

- how many?
- location
- types
- how to specify?

- instruction format

- size
- how many formats?

